

Applied Optimization - Ex 3

QCQP to SOCP

$$1.) (\Leftrightarrow) \left\| \frac{1 + (b^T x + c)}{Ax} \right\|_2 \leq \frac{1 - (b^T x + c)}{2}$$

$$\|Ax\|_2^2 + \left(\frac{1+u}{2}\right)^2 \leq \left(\frac{1-u}{2}\right)^2$$

$$\|Ax\|_2^2 + \frac{1+2u+u^2}{4} \leq \frac{1-2u+u^2}{4}$$

$$\|Ax\|_2^2 + \frac{4u}{4} \leq 0$$

$$x^T A^T A x + b^T x + c \leq 0$$

$$(\Rightarrow) x^T A^T A x + b^T x + c \leq 0$$

$$\|Ax\|_2^2 + u \leq 0$$

reverse algebra above

$$\left\| \frac{1 + (b^T x + c)}{Ax} \right\|_2 \leq \frac{1 - (b^T x + c)}{2}$$

$$2.) \text{ minimize } x_1^2 + 4x_1x_2 + 4x_2^2 = (x_1 + 2x_2)^2$$

$$\bullet \text{ s.t. } \begin{cases} 9x_1^2 + 16x_2^2 \leq 25 \\ x_1 - x_2 = 1 \end{cases} \text{ already in SOCP form}$$

$\Rightarrow$ epigraph transform

$$\begin{aligned} &\text{minimize } t \\ &\text{s.t. } x_1^2 + 4x_1x_2 + 4x_2^2 \leq t \quad (\text{quadratic inequality constraint}) \end{aligned}$$

$\Rightarrow$ algebra

$$x_1^2 + 4x_1x_2 + 4x_2^2 = x^T H x = x^T A^T A x = \|Ax\|_2^2$$

$\hookrightarrow$ for some A

$$\Rightarrow \|Ax\|_2 \leq \sqrt{t}$$

not affine

$$\Rightarrow \|Ax\|_2 \leq S, \quad S^2 \leq t$$

Not in SOCP form

$$\Rightarrow \|Ax\|_2 \leq S, \quad (2S, 1, t) \in Q_r^3$$

rotated second order cone

(let $u = b^T x + c$, and square both sides)

$$(\text{Note: } \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\|_2^2 = \left\| \begin{bmatrix} x_2 \\ x_1 \end{bmatrix} \right\|_2^2 + x_1^2)$$

Final problem

$$\begin{aligned} &\text{minimize } t \\ &\text{s.t. } \left\| \begin{bmatrix} 3x_1 \\ 4x_2 \end{bmatrix} \right\|_2 \leq 5 \end{aligned}$$

$$x_1 - x_2 = 1$$

$$\|Ax\|_2 \leq S$$

$$(2S, 1, t) \in Q_r^3$$

Linear Programming

$$\text{minimize } \|(2x_1 + 3x_2, -3x_1)^T\|_\infty$$

$$\text{s.t. } |x_1 - 2x_2| \leq 3$$

$$(1) \text{ Note } \|x\|_\infty = \max\{|x_1|, \dots, |x_n|\}$$

$$\text{minimize } t$$

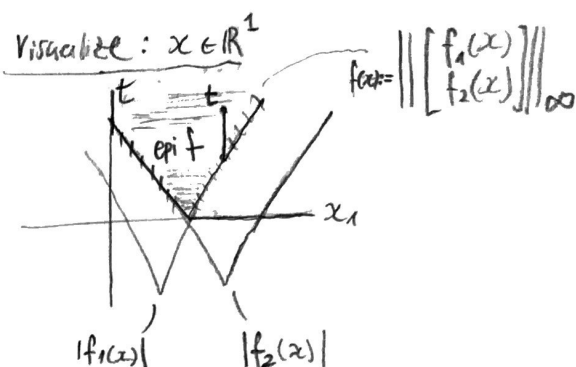
$$\text{s.t. } |2x_1 + 3x_2| \leq t$$

$$|-3x_1| \leq t$$

$$|x_1 - 2x_2| \leq 3$$

$$-(x_1 - 2x_2) \leq 3$$

epigraph transform.



$$\text{minimize } t$$

$$\text{s.t. } \pm(2x_1 + 3x_2) \leq t$$

$$\pm(-3x_1) \leq t$$

$$\pm(x_1 - 2x_2) \leq 3$$

6 constraints total (2 for each $\pm$)

(2) (a) All vars in $\mathbb{R}_+$

$$\text{minimize } t$$

$$\text{s.t. } \pm(2(x_1^+ - x_1^-) + 3(x_2^+ - x_2^-)) \leq t$$

$$\pm(-3(x_1^+ - x_1^-)) \leq t$$

$$\pm(x_1^+ - x_1^-) - 2(x_2^+ - x_2^-) \leq 3$$

$$x_1^+, x_1^-, x_2^+, x_2^- \geq 0$$

(b) Remove general inequality constraints

$$\text{minimize } t$$

$$x_1, t$$

$$\text{s.t. } 2(x_1^+ - x_1^-) + 3(x_2^+ - x_2^-) - t + s_1 = 0$$

$$-3(x_1^+ - x_1^-) - t + s_2 = 0$$

$$3(x_1^+ - x_1^-) - t + s_3 = 0$$

$$(x_1^+ - x_1^-) - 2(x_2^+ - x_2^-) - t + s_4 = 0$$

$$-(x_1^+ - x_1^-) - 2(x_2^+ - x_2^-) - t + s_5 = 0$$

$$-(x_1^+ - x_1^-) - 2(x_2^+ - x_2^-) - t + s_6 = 0$$

$$x_1^+, x_1^-, x_2^+, x_2^-, s_1, \dots, s_6 \geq 0$$

Also,

$$t = t^+ - t^-$$

$$t^+, t^- \geq 0$$

for all t above.

Transforming a general LP to standard form

Original problem

Let the general linear program be

$$\begin{aligned} & \text{minimize} && c^\top x + d \\ & \text{subject to} && Gx \preceq h \\ & && Ax = b \end{aligned}$$

where $G \in \mathbb{R}^{m \times n}$, $A \in \mathbb{R}^{p \times n}$, $c \in \mathbb{R}^n$, $d \in \mathbb{R}$, $h \in \mathbb{R}^m$, and $b \in \mathbb{R}^p$.

Step 1: Replace unrestricted variables by nonnegative pairs

We want to transform the problem to a form where all variables are positive or zero. For each component x_j introduce nonnegative variables $x_j^+ \geq 0$ and $x_j^- \geq 0$ and set

$$x_j = x_j^+ - x_j^-.$$

In vector form we define $x^+, x^- \in \mathbb{R}^n$ and thus write

$$x = x^+ - x^-, \quad x^+ \geq 0, \quad x^- \geq 0.$$

Step 2: Convert inequalities to equalities introducing slack variables

Introduce slack variables $s \in \mathbb{R}^m$ with $s \geq 0$ to convert $Gx \preceq h$ to an equality:

$$G(x^+ - x^-) + s = h, \quad s \geq 0.$$

Step 3: Stack variables and build standard-form system

Form the stacked decision vector

$$x' = \begin{bmatrix} x^+ \\ x^- \\ s \end{bmatrix} \in \mathbb{R}^{2n+m}, \quad x' \geq 0.$$

Define the block matrix B and right-hand side e by

$$B = \begin{bmatrix} G & -G & I_m \\ A & -A & 0_{p \times m} \end{bmatrix}, \quad e = \begin{bmatrix} h \\ b \end{bmatrix}.$$

Then the constraints become the single equality

$$Bx' = e.$$

Objective function in the new variables

The original objective transforms as

$$c^\top x + d = c^\top (x^+ - x^-) + d = p^\top x' + d, \quad p = \begin{bmatrix} c \\ -c \\ 0_m \end{bmatrix}.$$

where 0_m are m zeros in the vector. The constant d does not affect the minimization, so we can drop it.

Standard-form LP

Discarding the additive constant d in the optimization problem (it may be added back to report the original objective value), the equivalent standard-form LP is

$$\begin{aligned} & \text{minimize} && p^\top x' \\ & \text{subject to} && Bx' = e \\ & && x' \succeq 0 \end{aligned}$$

Relation between feasible sets, optimal solutions, and optimal values

Feasible sets. Let

$$F = \{x \in \mathbb{R}^n \mid Gx \preceq h, Ax = b\}$$

be the feasible set of the original LP and

$$F' = \{x' \in \mathbb{R}_+^{2n+m} \mid Bx' = e\}$$

the feasible set of the standard-form LP. Define the canonical mapping

$$\Phi: x \mapsto x' = (x^+, x^-, s) = (\max(x, 0), \max(-x, 0), h - Gx).$$

Then Φ maps F into F' . With the canonical choice $x^+ = \max(x, 0)$ and $x^- = \max(-x, 0)$ the map is bijective onto F' and invertible by $x = x^+ - x^-$. Thus feasibility is preserved exactly.

Optimal solutions. If x^* is optimal for the original LP then $x'^* = \Phi(x^*)$ is feasible for the standard form and attains

$$p^\top x'^* = c^\top x^*.$$

Conversely any optimal $x'^* \in F'$ yields $x^* = x'^* - x'^{-*} \in F$ with the same objective value (up to the constant d). Nonuniqueness in the split (x^+, x^-) can produce multiple x' corresponding to the same x if one does not fix the canonical split. Using the canonical split removes that ambiguity.

Optimal values. Let v_{orig} denote the optimal value of the original LP and v_{std} the optimal value of the standard-form LP with objective $p^\top x'$. Then

$$v_{\text{orig}} = v_{\text{std}} + d.$$

Therefore we can recover the optimal value easily.